

Algebraic Topology: a beginner's course

This lecture:

**AlgTop6: Non-orientable
surfaces--the Mobius band**

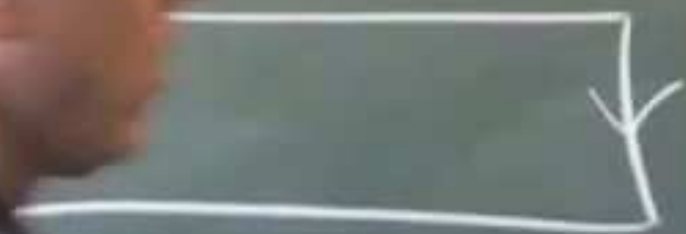
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Lecture 6: Non-orientable surfaces –
the Möbius band



Lecture 6: Non-orientable surfaces – the Möbius band

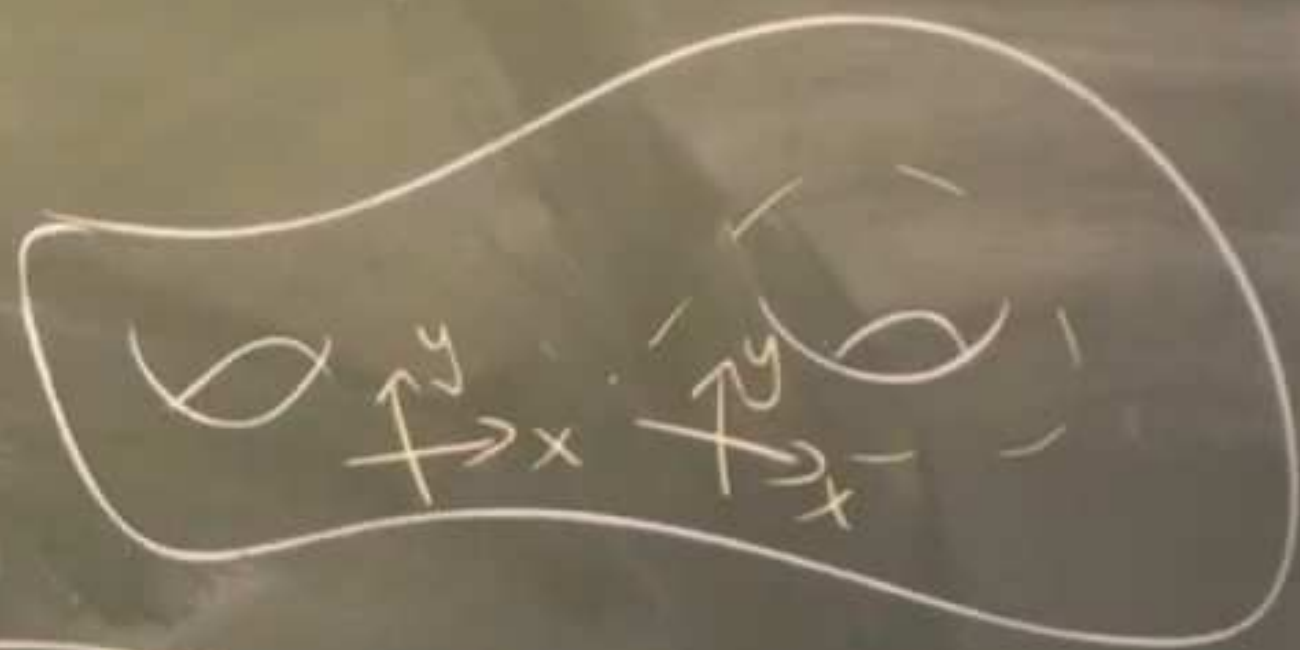
Möbius band



Alternate way:

y

x



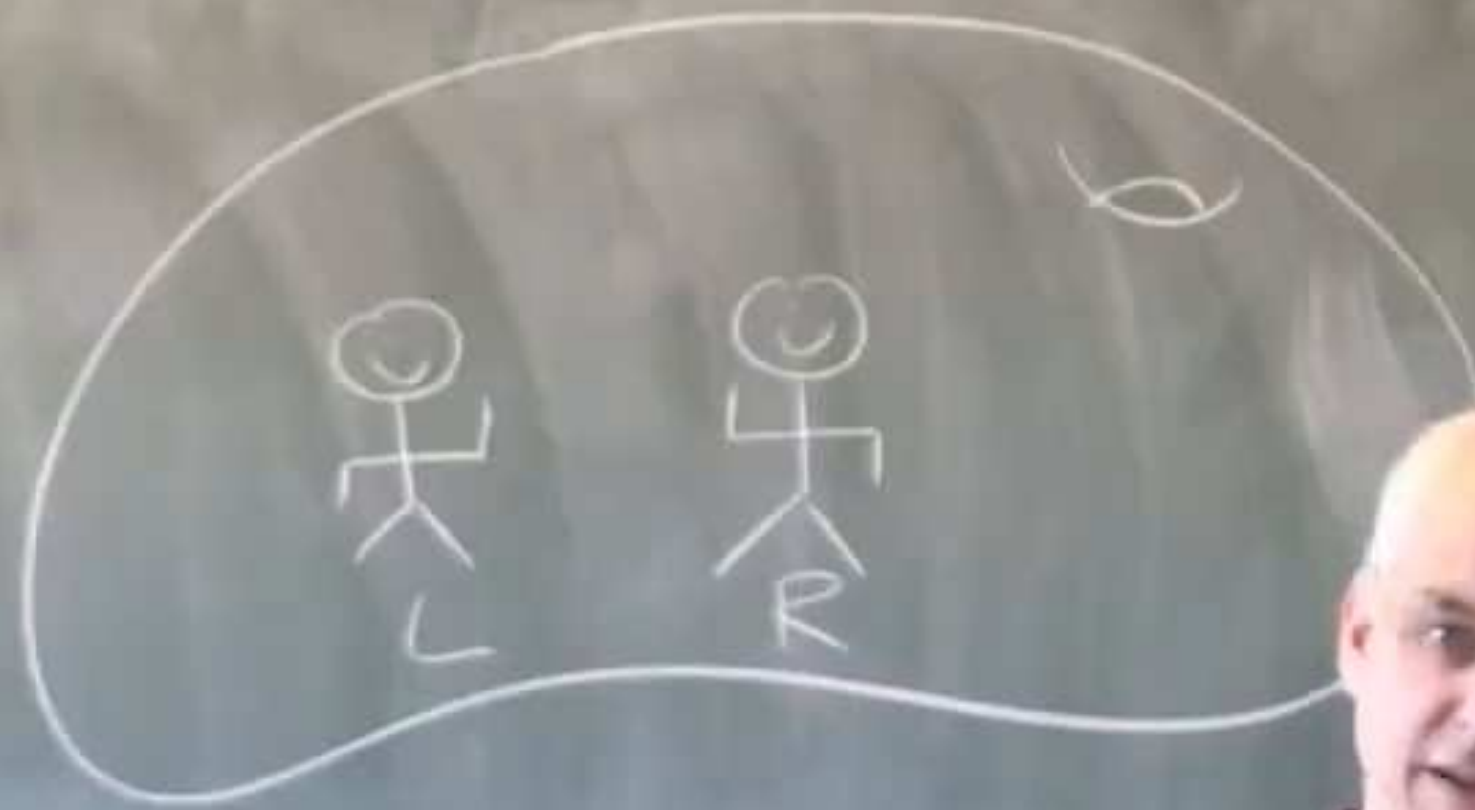
ble

ntly

rotation

orientable

orientable



Kleiv





Klein









$V_{OH} = 1$



Main thm Any 2-dim ^{compact} surface is a sphere
with a) a finite # of handles (ie sphere or n-holed)
or b) a finite # of crosscaps (ie projective plane)

1907 compact = closed + bounded



Main Thm Any 2-dim ^{compact} surface is a sphere
with a) a finite # of handles (ie sphere with holes)
or b) a finite # of crosscaps (ie projective plane)

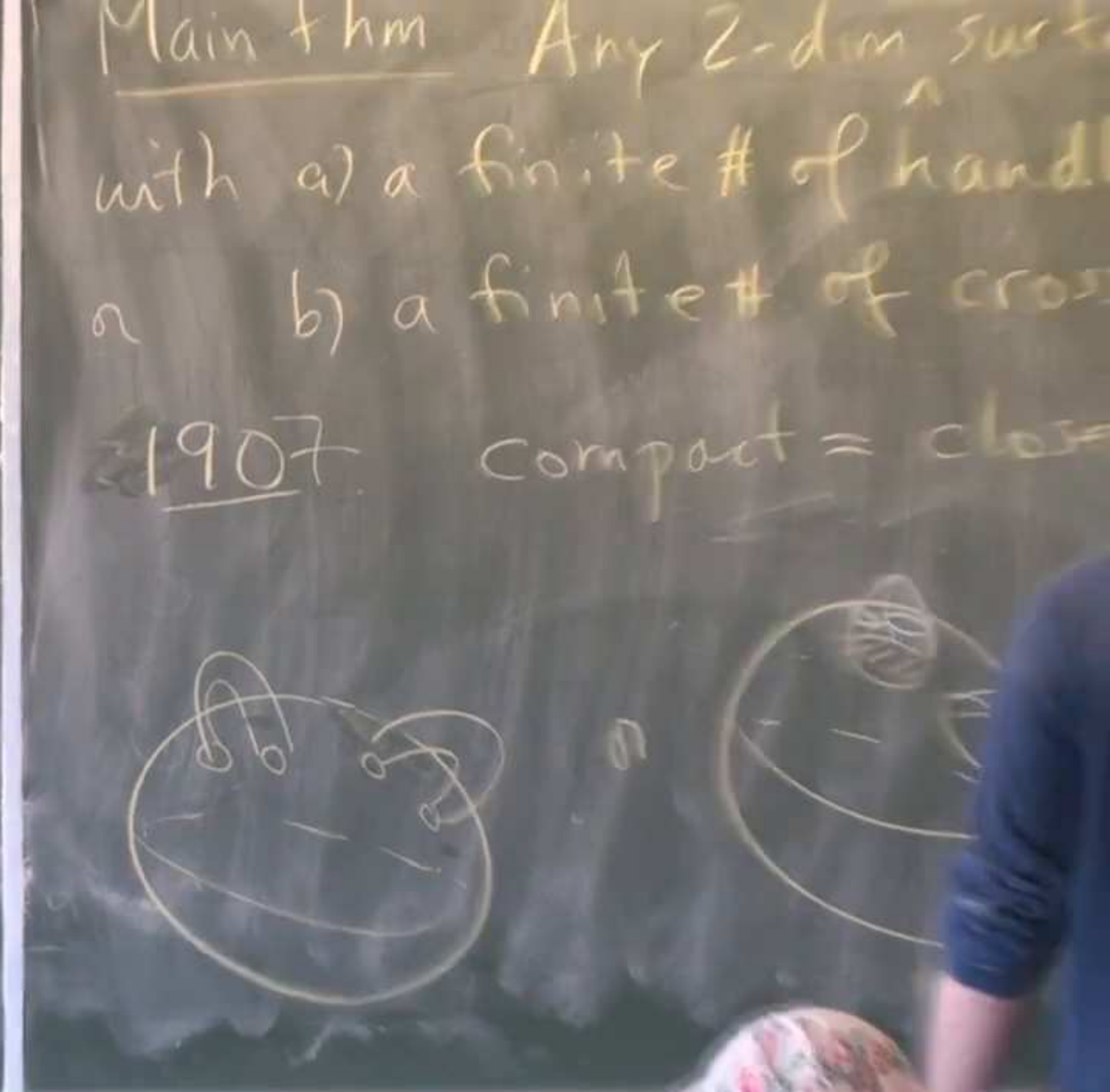
1907 compact = closed + bounded



Main thm Any 2-dim ^{compact} surface is a sphere
with a) a finite # of handles (ie sphere or n-holed torus)
or b) a finite # of crosscaps (ie. projective plane)

1907 compact = closed + bounded





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next in the series:

AlgTop7: The Klein bottle and projective plane

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